

**,VISVESVARAYA TECHNOLOGICAL
UNIVERSITY**

“JnanaSangama”, Belgaum -590014, Karnataka.



**LAB REPORT
on**

Machine Learning (23CS6PCMAL)

Submitted by

Rayhan (1BM23CS264)

in partial fulfillment for the award of the degree of

**BACHELOR OF ENGINEERING
in
COMPUTER SCIENCE AND ENGINEERING**



B.M.S. COLLEGE OF ENGINEERING

(Autonomous Institution under VTU)

BENGALURU-560019

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B.M.S. College of Engineering,
Bull Temple Road, Bangalore 560019
(Affiliated To Visvesvaraya Technological University, Belgaum)
Department of Computer Science and Engineering



CERTIFICATE

This is to certify that the Lab work entitled “Machine Learning (23CS6PCMAL)” carried out by **Rayhan Sadat (1BM23CS264)**, who is bonafide student of **B.M.S. College of Engineering**. It is in partial fulfillment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belgaum. The Lab report has been approved as it satisfies the academic requirements in respect of an Machine Learning (23CS6PCMAL) work prescribed for the said degree.

Lab faculty Incharge Name Assistant Professor Department of CSE, BMSCE	Dr. Kavitha Sooda Professor & HOD Department of CSE, BMSCE
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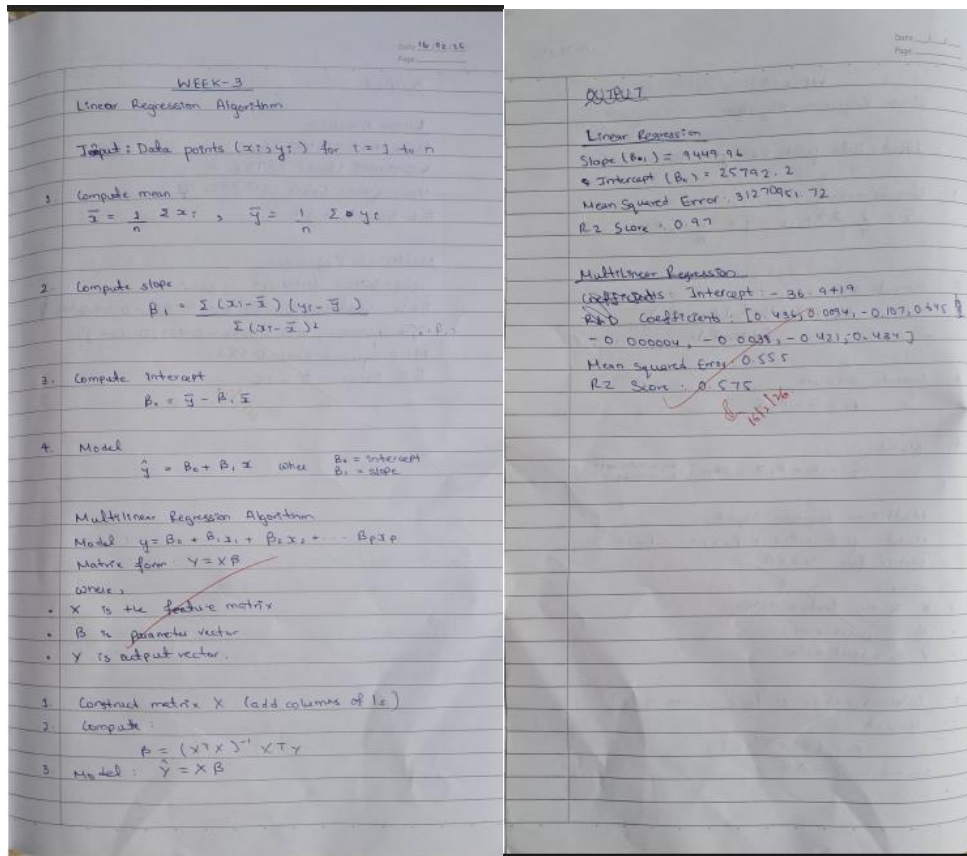
Sl. No.	Date	Experiment Title	Page No.
1	21-2-2025	Write a python program to import and export data using Pandas library functions	
2	3-3-2025	Demonstrate various data pre-processing techniques for a given dataset	
3	10-3-2025	Implement Linear and Multi-Linear Regression algorithm using appropriate dataset	
4	17-3-2025	Build Logistic Regression Model for a given dataset	
5	24-3-2025	Use an appropriate data set for building the decision tree (ID3) and apply this knowledge to classify a new sample.	
6	7-4-2025	Build KNN Classification model for a given dataset.	
7	21-4-2025	Build Support vector machine model for a given dataset	
8	5-5-2025	Implement Random forest ensemble method on a given dataset.	
9	5-5-2025	Implement Boosting ensemble method on a given dataset.	
10	12-5-2025	Build k-Means algorithm to cluster a set of data stored in a .CSV file.	
11	12-5-2025	Implement Dimensionality reduction using Principal Component Analysis (PCA) method.	

Github Link:

https://github.com/RayhanSadat26/ML_Lab

Program 3

Screenshot



Code:

```
from sklearn.datasets import fetch_california_housing
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
```

Load dataset

```
data = fetch_california_housing()
X = data.data[:, [0]] # Use only MedInc (first feature)
y = data.target
```

Split

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

Train

```
model = LinearRegression()
model.fit(X_train, y_train)
```

```
print("Intercept:", model.intercept_)
print("Slope:", model.coef_[0])
```

```
print("R^2:", model.score(X_test, y_test))
```

O/P:

Intercept: 0.4445972916907872

Slope: 0.4193384939381273

R^2: 0.45885918903846656

```
from sklearn.datasets import fetch_california_housing
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
```

```
# Load dataset
```

```
data = fetch_california_housing()
```

```
X = data.data # All features
```

```
y = data.target
```

```
# Split
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

```
# Train
```

```
model = LinearRegression()
```

```
model.fit(X_train, y_train)
```

```
print("Intercept:", model.intercept_)
```

```
print("Coefficients:", model.coef_)
```

```
print("R^2:", model.score(X_test, y_test))
```

O/P:

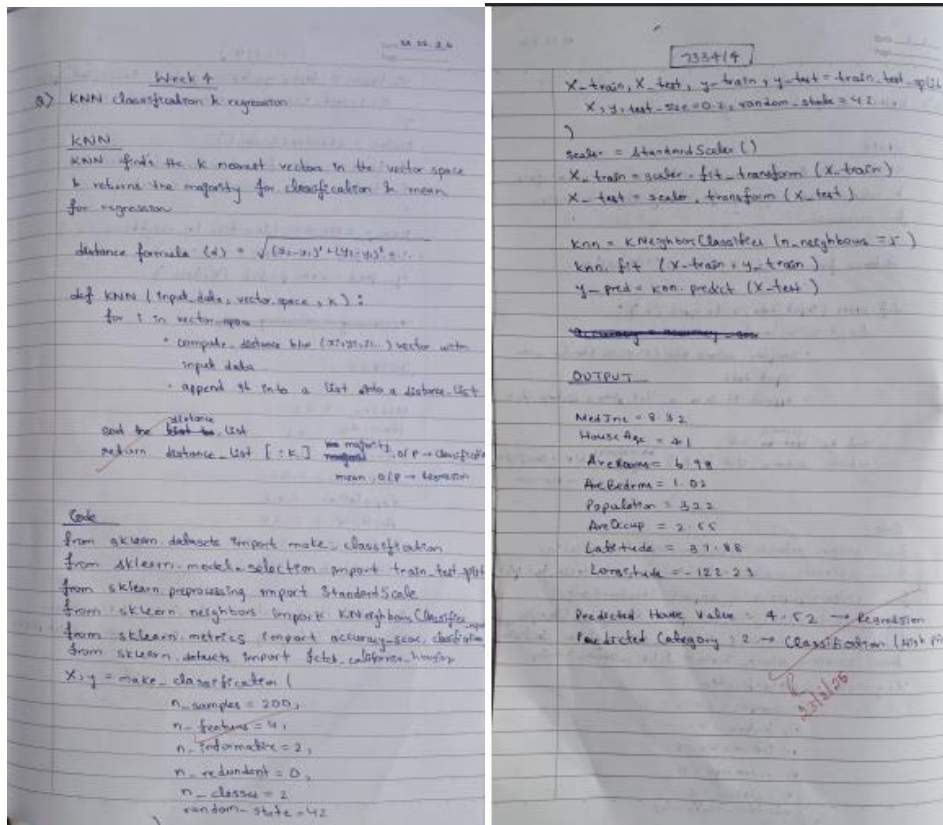
Intercept: -37.023277706063794

Coefficients: [4.48674910e-01 9.72425752e-03 -1.23323343e-01 7.83144907e-01
-2.02962058e-06 -3.52631849e-03 -4.19792487e-01 -4.33708065e-01]

R^2: 0.5757877060324524

Program 4

Screenshot



Code:

import numpy as np

from sklearn.neighbors **import** KNeighborsClassifier, KNeighborsRegressor

X = np.array([[100], [105], [98], [150], [160], [155]])

y_class = np.array([0, 0, 0, 1, 1, 1])

y_reg = np.array([100, 105, 98, 150, 160, 155])

1) KNN Classification

knn_classifier = KNeighborsClassifier(n_neighbors=3)

knn_classifier.fit(X, y_class)

new_ball = np.array([[110]])

```
class_prediction = knn_classifier.predict(new_ball)

if class_prediction[0] == 0:

    print("Classification: Red Ball")

else:

    print("Classification: Blue Ball")

# 2) KNN Regression

knn_regressor = KNeighborsRegressor(n_neighbors=3)

knn_regressor.fit(X, y_reg)

weight_prediction = knn_regressor.predict(new_ball)

print("Regression (Average Predicted Weight):", weight_prediction[0])
```

O/P:

Classification: Red Ball

Regression (Average Predicted Weight): 101.0